Sensitivity Analysis using Linear Programming

Detailed Presentation for Final Project

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UGST4090: Introduction to Optimization

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What is Sensitivity Analysis?

Sensitivity analysis studies how changes in the parameters of a linear program affect:

- the **optimal solution**
- the **optimal objective value**
- the **stability of the current basis**

Main question: if an LP parameter changes, does the current solution remain optimal?

Why it matters:

- real data are uncertain
- costs, profits, and resource availability change
- decision-makers need to know when a model is robust

Core Logic of Sensitivity Analysis

For a **maximization** problem, the current basic feasible solution remains optimal if:

1. the tableau still satisfies the **feasibility condition** (right-hand side values remain nonnegative)
2. the tableau still satisfies the **optimality condition** (reduced costs remain nonnegative)

If one of these conditions fails:

- negative RHS entry $\Rightarrow$ current basis becomes **infeasible**
- negative reduced cost $\Rightarrow$ current basis becomes **suboptimal**

Radioco Radio Production Problem

Radioco manufactures two types of radios and wants to maximize weekly profit.

Item	**Radio 1**	**Radio 2**
Price ($)	25	22
Laborer 1 hours	1	2
Laborer 2 hours	2	1
Raw material cost ($)	5	4

Let:

$$x_1 = \text{number of type-1 radios} \qquad x_2 = \text{number of type-2 radios}$$

LP Formulation

The model is:

$$\max z = 3x_1 + 2x_2$$

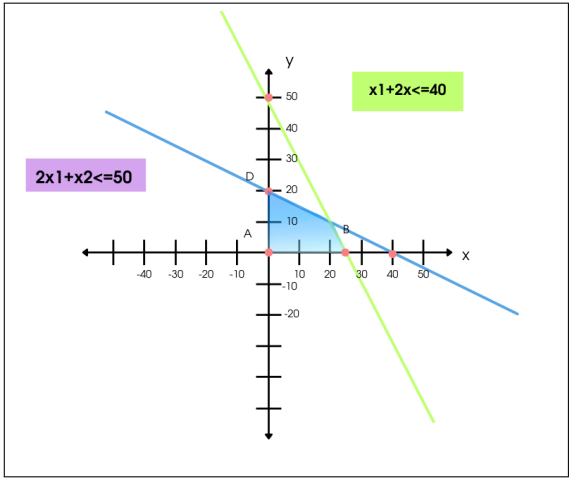
subject to

$$\begin{aligned} x_1 + 2x_2 &\leq 40 \quad \text{(Laborer 1)} \\ 2x_1 + x_2 &\leq 50 \quad \text{(Laborer 2)} \\ x_1, x_2 &\geq 0 \end{aligned}$$

Interpretation

- each constraint limits available labor hours
- feasible solutions lie in the intersection of all constraints
- the objective selects the best feasible production mix

Feasible Region and Optimal Point



Visualizing Constraints and Feasible Region

At the optimum:

$$x_1 = 20, \quad x_2 = 10, \quad z = 80$$

What this means:

- produce 20 units of Radio 1 and 10 units of Radio 2
- weekly profit equals 80
- both constraints are **binding**

Key insight: Small changes around a binding optimum directly affect the solution

Six Common Types of LP Changes

We ask the same question:

Does the current basis remain feasible and optimal after the modification?

Six changes:

1. Changing the objective coefficient of a **nonbasic variable**
2. Changing the objective coefficient of a **basic variable**
3. Changing the **right-hand side** of a constraint
4. Changing the **column of a nonbasic variable**
5. **Adding a new activity** (new variable)
6. **Adding a new constraint**

Cases 1 & 2 – Objective function coefficients

1. Objective coefficient of a nonbasic variable
   - feasibility $\rightarrow$ unchanged
   - optimality $\rightarrow$ coefficient nonnegative
2. Objective coefficient of a basic variable
   - feasibility $\rightarrow$ unchanged
   - optimality $\rightarrow$ recompute

Changing an Objective Function Coefficient

The new objective becomes

$$\max z = c_1 x_1 + 2x_2$$

Each isoprofit line satisfies

$$c_1 x_1 + 2x_2 = k \quad \Rightarrow \quad x_2 = -\frac{c_1}{2} x_1 + \frac{k}{2}$$

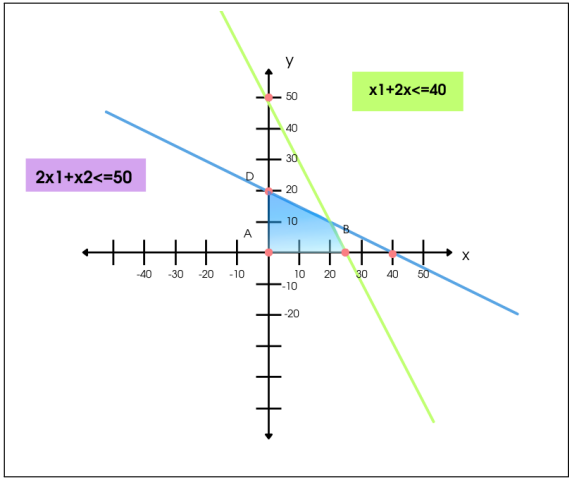
So the slope of the objective line is

$$-\frac{c_1}{2}$$

Constraint slopes are:

$$x_1 + 2x_2 = 40 \Rightarrow -\frac{1}{2}, \qquad 2x_1 + x_2 = 50 \Rightarrow -2$$

Changing an Objective Function Coefficient



Visualizing Constraints and Feasible Region

To remain optimal:

$$-2 \leq -\frac{c_1}{2} \leq -\frac{1}{2}$$

Solving gives:

$$1 \leq c_1 \leq 4$$

Case 3 - Changing a Right-Hand Side Value

3. Right-hand side of a constraint
   - feasibility $\rightarrow$ can change
   - optimality $\rightarrow$ unchanged

Now change the first RHS from 40 to $40 + \Delta$:

$$x_1 + 2x_2 = 40 + \Delta, \qquad 2x_1 + x_2 = 50$$

Solving the system yields:

$$x_1 = 20 + \frac{\Delta}{3}, \qquad x_2 = 10 - \frac{\Delta}{3}$$

Shadow Price

Substitute the adjusted solution into the objective:

$$z = 3x_1 + 2x_2 = 3\left(20 + \frac{\Delta}{3}\right) + 2\left(10 - \frac{\Delta}{3}\right)$$

Thus,

$$z = 80 + \frac{\Delta}{3}$$

So a 1-unit increase in the first RHS raises the objective by:

$$\frac{1}{3}$$

Shadow price of Constraint 1:

$$\pi_1 = \frac{1}{3}$$

Cases 4–6: What Happens?

4. Changing the column of a nonbasic variable
   - feasibility $\rightarrow$ unchanged
   - optimality $\rightarrow$ coefficient nonnegative

5. Adding a new activity
   - feasibility $\rightarrow$ unchanged
   - optimality $\rightarrow$ coefficient nonnegative

6. Adding a new constraint
   - Case 1 – Optimal
   - Case 2 – Feasible
   - Case 3 – Infeasible

But why do we care about all those methods?

Sensitivity analysis is about figuring out whether solution is still optimal after the changes to initial setup

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But why not use something simpler?

How else can we verify what the change does? — just resolve!

Resolving!

Alot of code runs that compare complete resolving vs usage of sensitivity

C2/C3 – around x200 speedup
C1/C4/C5/C6 – around x600 speedup, scales with the system size

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So how is speedup like that achieved?

Clever Reuse

Suppose, that we are back at the step one. We have initial tableau (updated this time). Can we obtain the solution, if now we additionally know which variables will end up as basic?

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Yes, we can!

Clever Reuse

BVs are x2 and s2

z	x_1	x_2	s_1	s_2	rhs	Basic Variable
1	-2	3	0	0	0	$z = 0$
0	1	1	1	0	4	$s_1 = 4$
0	1	-1	0	1	6	$s_2 = 6$

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Derivations from the book:

x_j column in optimal tableau's constraints $= B^{-1}\mathbf{a}_j$

Right-hand side of optimal tableau's constraints $= B^{-1}\mathbf{b}$

$\bar{c}_j = \mathbf{c}_{\mathrm{BV}} B^{-1}\mathbf{a}_j - c_j$

Coefficient of slack variable s_i in optimal row 0

$= i$th element of $\mathbf{c}_{\mathrm{BV}} B^{-1}$

Coefficient of excess variable e_i in optimal row 0

$= -(i$th element of $\mathbf{c}_{\mathrm{BV}} B^{-1})$

Coefficient of artificial variable a_i in optimal row 0

$= (i$th element of $\mathbf{c}_{\mathrm{BV}} B^{-1}) + M$ (max problem)

Right-hand side of optimal row 0 $= \mathbf{c}_{\mathrm{BV}} B^{-1}\mathbf{b}$

Clever Reuse

But matrices are very expensive to invert! $O(n^3)$ to be precise

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But matrices are very expensive to invert! $O(n^3)$ to be precise
....Except we have done it already.

$$B|I_3 = \left[\begin{array}{ccc|ccc} 1 & 1 & 8 & 1 & 0 & 0 \\ 0 & 1.5 & 4 & 0 & 1 & 0 \\ 0 & 0.5 & 2 & 0 & 0 & 1 \end{array}\right]$$

Bit of History

Fun fact: First version of the simplex algorithm was published 1947. 6 years later, its author published the updatd version (Revised simplex method) which by that time already used $B^{-}1$ in some way.

George Dantzig

Finance Application

Sensitivity analysis is widely used in valuation models to test how results respond to uncertain assumptions

For instance, **price per share** calculation based on a Discounted Cash Flow (DCF) framework

Two key inputs are stressed:

- **WACC** (Weighted Average Cost of Capital)
- **Terminal Growth Rate**

Baseline assumptions from the report:

$$\text{Price per Share} = 118.0, \qquad \text{Terminal Growth} = 1.25\%, \qquad \text{WACC} = 10.63\%$$

Sensitivity Analysis: Interpretation

Terminal Growth Rate

	1.00%	1.25%	1.50%	1.75%	2.00%
WACC					
9.50%	135.4	139.1	142.9	147.1	151.4
10.00%	126.2	129.4	132.8	136.3	140.1
10.50%	118.0	120.8	123.7	126.9	130.2
11.00%	110.5	113.0	115.6	118.4	121.3
11.50%	104.0	106.4	108.9	111.6	114.5

(rows = WACC, columns = growth)

How to read:

- Cells: price per share
- Rows: WACC
- Columns: growth

Key insights:

- Price ranges from **104 to 151**
- Higher WACC $\rightarrow$ lower value
- Higher growth $\rightarrow$ higher value

Takeaway:

Result depends on assumptions $\Rightarrow$ uncertainty matters

Conclusion

- Sensitivity analysis tells us whether the current LP solution remains valid after parameter changes.
- In the Radioco example, we examined both coefficient changes and RHS changes graphically.
- The shadow price gives the marginal value of relaxing a binding resource constraint.
- In simplex terms, the current basis remains optimal only when feasibility and optimality conditions continue to hold.
- In finance, sensitivity analysis helps evaluate uncertainty in valuation models such as DCF.